

# Muhammad Rafay Irfan

Software Engineer · Python · C++ · C · AWS · End-to-End Product Delivery

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## Work Experience

### Google Summer of Code (lib2geom, Inkscape)

May 2025 – Sep. 2025

Open Source Developer

Remote

- Owned a geometry-library deliverable end-to-end in C++ – scoping the problem from first principles, selecting the numerical methods, and shipping a 50+ case Google Test suite covering tangent, overlapping, and precision-sensitive edge cases – merged into a production codebase used by millions.
- Wrote high-quality, maintainable C++ to resolve a long-standing robustness gap in lib2geom's intersection engine, replacing fragile floating-point logic with error-bounded interval arithmetic and fixing incorrect results on degenerate curve-curve configurations.
- Worked directly with senior open-source maintainers through async code review cycles, refining API surface design and edge-case handling to the standards of a mature geometry library.

### Inkscape

Oct. 2024 – May 2025

Open Source Developer

Remote

- Independently clarified ambiguous technical requirements by reading unfamiliar code and writing minimal reproducible test cases, in a multi-hundred-thousand-line C++/GTK project with no onboarding documentation.
- Diagnosed and resolved high-impact bugs across rendering, tool interaction, and editor-state layers, shipping targeted, regression-free fixes for 3M+ active users.
- Maintained CI/CD pipeline integrity across 5+ dependent tool behaviors throughout the project's SDLC, and communicated technical decisions clearly in async review threads without direct mentorship.

### Systems Limited

Jul. 2023 – Sep. 2023 & Jul. 2024 – Sep. 2024

Software Engineering Intern

Lahore, Pakistan

- Designed and deployed cloud solutions on AWS (EC2, S3, RDS, Lambda) across 3 production environments, building serverless, event-driven functions to support enterprise application workflows.
- Strengthened environment security by implementing IAM policies and bucket access controls, reducing unauthorized-access risk across all 3 environments.
- Collaborated with the Cloud App Development team, gaining hands-on training in Java and contributing to enterprise application development practices.

## Projects

### Real-Time Healthcare AI Voice Agent | Node.js, TypeScript, Twilio, AWS Bedrock, ECS/Fargate, PostgreSQL, S3

- Shipped a production-quality, HIPAA-eligible real-time voice system integrating Twilio Media Streams with AWS Bedrock Nova Sonic, owning the architecture end-to-end through a KMS-encrypted S3 pipeline, PostgreSQL audit trails, and a full integration test suite.
- Built a config-driven, cloud-native multi-agent architecture that dynamically loads each agent's prompt, voice, locale, and call rules, supporting 20 distinct agents without code duplication.

### SmartEngage | Python, FastAPI, XGBoost, PyTorch, LangChain, PostgreSQL, Docker, ChromaDB

- Designed modular, reusable FastAPI service layers that ingest social media telemetry via async pipelines and predict content virality using a hybrid XGBoost + transformer model.
- Built an agentic RAG loop backed by a vector database and LLM orchestrator to route underperforming drafts for revision, containerized end-to-end with Docker.

### Vigil | Python, FastAPI, PostgreSQL, SQLAlchemy (async), Alpaca API, Pydantic, pytest/hypothesis

- Built an autonomous options-trading agent for the lablab.ai × Alpaca AI Trading Agents Hackathon, splitting responsibilities between a deterministic strategy/risk engine and an LLM portfolio manager so no model output reaches the broker unchecked.
- Designed a 12-gate deterministic risk kernel (per-trade and drawdown limits, concentration, portfolio delta, liquidity, credit quality, event blackout, idempotency) enforced as pure functions ahead of every order, with a Hypothesis property test proving the per-trade risk invariant across generated inputs and a 241-test suite passing under mypy --strict.
- Implemented multi-leg order execution and position management on Alpaca (limit-only, price-laddered), an async PostgreSQL/SQLAlchemy persistence layer with Decimal-based money handling, and a fixed in-process cron scheduler for entry, management, and end-of-day flatten cycles.

## Education

Ghulam Ishaq Khan Institute

2022 – 2026

BS. Computer Science

## Skills

**Languages:** C++, Python, TypeScript, JavaScript, Java, C **Frontend:** React, Next.js, Tailwind CSS

**Backend & APIs:** FastAPI, Node.js, REST API Design, Microservices, Docker, Kubernetes (familiar) **Databases:** PostgreSQL, MySQL, MongoDB, ChromaDB

**ML / AI:** PyTorch, XGBoost, Scikit-learn, Transformers, RAG, LangChain, LLMs, Pandas, NumPy

**Cloud & Infra:** AWS (EC2, S3, RDS, Lambda, ECS/Fargate, Bedrock) **Tools:** Git, CMake, Google Test, CI/CD, Linux